WEEK 7: SQL QUERY IMPLEMENTATION FOR E-COMMERCE DATABASE

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Scenario:

The Art Workshop maintains information about product,customer,order,and inventory data.The management wants to retrieve useful information from the database for daily operations and basic business analysis. We will be applying SQL queries to search,filter,sort,and generate reports from the existing Art Workshop database.

1.Perform Basic SQL Queries

SELECT Command is used to retrieve data from database tables

- It displays all Customer details,product details and Price details.

TASKS:

1.Display all Customer details.

```
SQL> SELECT * FROM Customer;

CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                                PHONE CREATED_A
------------------------------ ---------- ---------
C-1001               Beema                ponting
beema200@edss.com                9234567856 14-FEB-23

C-1002               priya                sakthi
priyasakthi@edss.com             9856734456 23-SEP-24

C-1003               Arun                 kumar
arun200@edss.com                 9876543212 15-OCT-24


CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                                PHONE CREATED_A
------------------------------ ---------- ---------
C-1004               Anu                  Priya
anupriya200@edss.com             8765943256 23-MAY-25

C-1005               Reena                das
reenadas200@edss.com             9087654567 30-APR-22

C-1006               Kaviya               Suresh
kaviya106@gmail.com              9876543210 10-JUN-25


CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                                PHONE CREATED_A
------------------------------ ---------- ---------
C-1007               Rahul                Kumar
rahul107@gmail.com               9876543211 15-JUL-25

C-1008               Divya                Rani
```

```
C-1009                Sanjay               Raj
sanjay109@gmail.com             9876543213 05-SEP-25


CUSTOMER_ID           FIRST_NAME           LAST_NAME
--------------------- -------------------- --------------------
EMAIL                                   PHONE CREATED_A
------------------------------ ---------- ---------
C-1010                Priya                Devi
priya110@gmail.com              9876543214 12-OCT-25

C-1011                Vijay                Kumar
vijay111@gmail.com              9876543215 18-NOV-25

C-1012                Meena                Lakshmi
meena112@gmail.com              9876543216 22-DEC-25


CUSTOMER_ID           FIRST_NAME           LAST_NAME
--------------------- -------------------- --------------------
EMAIL                                   PHONE CREATED_A
------------------------------ ---------- ---------
C-1013                Arjun                Raj
arjun113@gmail.com              9876543217 14-JAN-26

C-1014                Sneha                Rani
sneha114@gmail.com              9876543218 09-FEB-26

C-1015                Nithin               Kumar
nithin115@gmail.com             9876543219 16-MAR-26


15 rows selected.
```

2.Display all available products.

```
SQL> SELECT * FROM Product;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
P1001
Laptops
SEM0001      67500              2

P1002
Croptops and shirts
SEM0002      45000            456

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------

P1003
Fish tank
SEM0003       3400              4

P1004
Television

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
SEM0004      75000              5

P1005
```

```
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGOR       PRICE STOCK_QUANTITY
------- ----------- ---------------
SEM0004       75000                5

P1005
Unruled notes
SEM0005        7500              765

P1006

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGOR       PRICE STOCK_QUANTITY
------- ----------- ---------------
Tiles
SEM0006       55000              670
```

```
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
SEM0004      75000              5

P1005
Unruled notes
SEM0005       7500            765

P1006

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
Tiles
SEM0006      55000            670

P1007
Headphones
SEM0001       2500             25


PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
P1008
Keyboard
SEM0001       1500             30

P1009
T-Shirts
SEM0002        800             40
```

```
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
P1015
Rice Bag
SEM0006       1200             50


15 rows selected.
```

3.Retrieve product names and prices only.

```
SQL> SELECT PRODUCT_NAME, PRICE
  2  FROM Product;

PRODUCT_NAME
----------------------------------------
     PRICE
----------
Laptops
     67500

Croptops and shirts
     45000

Fish tank
      3400


PRODUCT_NAME
----------------------------------------
     PRICE
----------
Television
     75000

Unruled notes
      7500

Tiles
     55000


PRODUCT_NAME
----------------------------------------
     PRICE
----------
Headphones
      2500

Keyboard
      1500
```

```
T-Shirts
        800


PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
Jeans
       1800

Notebooks
        100

Pens
         20


PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
Mixer Grinder
       3500

Mobile Charger
        600

Rice Bag
       1200


15 rows selected.
```

4.Display all orders placed by customers.

```
SQL> SELECT CONCAT(c.FIRST_NAME, c.LAST_NAME) AS CUSTOMER_NAME,
  2         o.ORDER_ID,
  3         o.TOTAL_AMOUNT,
  4         o.ORDER_STATUS
  5  FROM Customer c
  6  JOIN Orders o
  7  ON c.CUSTOMER_ID = o.CUSTOMER_ID;

CUSTOMER_NAME                          ORDER_ID TOTAL_AMOUNT
-------------------------------------- ---------- ------------
ORDER_STATUS
--------------------
Beemaponting                                1001        70000
Pending

priyasakthi                                 1002        45000
Shipped

Arunkumar                                   1003         3400
Delivered


CUSTOMER_NAME                          ORDER_ID TOTAL_AMOUNT
-------------------------------------- ---------- ------------
ORDER_STATUS
--------------------
AnuPriya                                    1004        75000
Pending

Reenadas                                    1011        60000
Pending

Beemaponting                                1006        55000
Delivered


CUSTOMER_NAME                          ORDER_ID TOTAL_AMOUNT
-------------------------------------- ---------- ------------
ORDER_STATUS
```

```
CUSTOMER_NAME                                ORDER_ID TOTAL_AMOUNT
-------------------------------------------- ---------- ------------
ORDER_STATUS
--------------------
priyasakthi                                        1007        67500
Pending

Arunkumar                                          1008        45000
Shipped

AnuPriya                                           1009         3400
Delivered


CUSTOMER_NAME                                ORDER_ID TOTAL_AMOUNT
-------------------------------------------- ---------- ------------
ORDER_STATUS
--------------------
Reenadas                                           1010         7500
Pending


10 rows selected.
```

5.Retrieve payment details.

```
SQL> SELECT * FROM Payment;

PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
         1       1001 20-SEP-26 UPI                  Success
             70000

         2       1002 21-SEP-26 Card                 Success
             45000

         3       1003 22-SEP-26 Cash                 Success
              3400

PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
         4       1004 23-SEP-26 UPI                  Pending
             75000

         5       1006 25-SEP-26 Card                 Success
             55000

         6       1007 26-SEP-26 UPI                  Pending
             67500

PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
         7       1008 27-SEP-26 Card                 Success
             45000

         8       1009 28-SEP-26 Cash                 Success
              3400
```

```
         8       1009 28-SEP-26 Cash                Success
              3400

        10       1011 30-SEP-26 UPI                 Success
             60000


9 rows selected.
```

2.Apply Filtering Conditions Using WHERE:

TASKS: 1.Find Products with price greater than ₹5000

```
SQL> SELECT * FROM Product WHERE PRICE > 5000;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
P1001
Laptops
SEM0001      67500              2

P1002
Croptops and shirts
SEM0002      45000            456

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------

P1004
Television
SEM0004      75000              5

P1005
Unruled notes

PRODUCT
-------
```

```
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
SEM0005        7500            765

P1006
Tiles
SEM0006       55000            670
```

2.Display products available in stock.

```
SQL> SELECT * FROM Inventory WHERE Stock_Status = 'Available';

INVENTO PRODUCT SELLER_ STOCK_QUANTITY STOCK_STATU LAST_UPDA
------- ------- ------- -------------- ----------- ---------
Inv0001 P1001   S00001               4 Available   12-SEP-26
Inv0002 P1002   S00002               3 Available   13-SEP-26
Inv0004 P1004   S00004               3 Available   14-SEP-26
Inv0005 P1005   S00005              10 Available   16-SEP-26
Inv0006 P1006   S00006              12 Available   17-SEP-26
Inv0007 P1007   S00001              15 Available   18-SEP-26
Inv0008 P1008   S00002              20 Available   19-SEP-26
Inv0009 P1009   S00003               8 Available   20-SEP-26
Inv0010 P1010   S00004               6 Available   21-SEP-26
Inv0011 P1011   S00005              25 Available   22-SEP-26
Inv0012 P1012   S00006              30 Available   23-SEP-26

INVENTO PRODUCT SELLER_ STOCK_QUANTITY STOCK_STATU LAST_UPDA
------- ------- ------- -------------- ----------- ---------
Inv0013 P1013   S00001               7 Available   24-SEP-26
Inv0014 P1014   S00002               9 Available   25-SEP-26
Inv0015 P1015   S00003              14 Available   26-SEP-26

14 rows selected.
```

3.Display completed orders.

```
SQL> SELECT * FROM Orders WHERE ORDER_STATUS = 'Delivered';

  ORDER_ID CUSTOMER_ID         ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- ------------------- --------- ------------ -------------------
      1003 C-1003              22-SEP-26         3400 Delivered
      1006 C-1001              25-SEP-26        55000 Delivered
      1009 C-1004              28-SEP-26         3400 Delivered
```

4.Find products with rating above 4.

```
SQL> SELECT P.PRODUCT_NAME, ROUND(AVG(R.RATING), 2) AS AVERAGE_RATING FROM Product P JOIN Review R ON P.PRODUC
T_ID = R.PRODUCT_ID GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME HAVING AVG(R.RATING) > 4 ORDER BY AVERAGE_RATING DES
C;

PRODUCT_NAME
--------------------------------------------------------------------------------
AVERAGE_RATING
--------------
Unruled notes
            5

Laptops
          4.5
```

3.Sort Data Using Order by

TASK:

1.Display products from lowest to highest price.

```
SQL> SELECT * FROM Product ORDER BY Price;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
CATEGOR       PRICE STOCK_QUANTITY
------- ---------- --------------
P1012
Pens
SEM0003          20            200

P1011
Notebooks
SEM0003         100            100

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
CATEGOR       PRICE STOCK_QUANTITY
------- ---------- --------------

P1014
Mobile Charger
SEM0005         600             35

P1009
T-Shirts

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
CATEGOR       PRICE STOCK_QUANTITY
------- ---------- --------------
SEM0002         800             40
```

```
P1015
Rice Bag
SEM0006         1200                50

P1008

PRODUCT
-------
PRODUCT_NAME
----------------------------------------
CATEGOR       PRICE STOCK_QUANTITY
------- ---------- ----------------
Keyboard
SEM0001         1500                30

P1010
Jeans
SEM0002         1800                20


PRODUCT
-------
PRODUCT_NAME
----------------------------------------
CATEGOR       PRICE STOCK_QUANTITY
------- ---------- ----------------
P1007
Headphones
SEM0001         2500                25

P1003
Fish tank
SEM0003         3400                 4

PRODUCT
-------
PRODUCT_NAME
----------------------------------------
```

```
P1013
Mixer Grinder
SEM0004         3500                15

P1005
Unruled notes

PRODUCT
-------
PRODUCT_NAME
----------------------------------------
CATEGOR       PRICE STOCK_QUANTITY
------- ---------- ----------------
SEM0005          7500              765

P1002
Croptops and shirts
SEM0002        45000              456

P1006

PRODUCT
-------
PRODUCT_NAME
----------------------------------------
CATEGOR       PRICE STOCK_QUANTITY
------- ---------- ----------------
Tiles
SEM0006        55000              670

P1001
Laptops
SEM0001        67500                2
```

```
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
P1004
Television
SEM0004      75000              5


15 rows selected.
```

2.Display customers alphabetically.

```
SQL> SELECT * FROM Customer ORDER BY First_Name;

CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                                  PHONE CREATED_A
------------------------------ ---------- ---------
C-1004               Anu                  Priya
anupriya200@edss.com             8765943256 23-MAY-25

C-1013               Arjun                Raj
arjun113@gmail.com               9876543217 14-JAN-26

C-1003               Arun                 kumar
arun200@edss.com                 9876543212 15-OCT-24


CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                                  PHONE CREATED_A
------------------------------ ---------- ---------
C-1001               Beema                ponting
beema200@edss.com                9234567856 14-FEB-23

C-1008               Divya                Rani
divya108@gmail.com               9876543212 20-AUG-25

C-1006               Kaviya               Suresh
kaviya106@gmail.com              9876543210 10-JUN-25


CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                                  PHONE CREATED_A
------------------------------ ---------- ---------
C-1012               Meena                Lakshmi
meena112@gmail.com               9876543216 22-DEC-25
```

```
C-1015                    Nithin                    Kumar
nithin115@gmail.com                 9876543219 16-MAR-26

C-1010                    Priya                     Devi
priya110@gmail.com                  9876543214 12-OCT-25


CUSTOMER_ID               FIRST_NAME                LAST_NAME
------------------------- ------------------------- -------------------
EMAIL                                       PHONE CREATED_A
------------------------------ ---------- ---------
C-1007                    Rahul                     Kumar
rahul107@gmail.com                  9876543211 15-JUL-25

C-1005                    Reena                     das
reenadas200@edss.com                9087654567 30-APR-22

C-1009                    Sanjay                    Raj
sanjay109@gmail.com                 9876543213 05-SEP-25


CUSTOMER_ID               FIRST_NAME                LAST_NAME
------------------------- ------------------------- -------------------
EMAIL                                       PHONE CREATED_A
------------------------------ ---------- ---------
C-1014                    Sneha                     Rani
sneha114@gmail.com                  9876543218 09-FEB-26

C-1011                    Vijay                     Kumar
vijay111@gmail.com                  9876543215 18-NOV-25

C-1002                    priya                     sakthi
priyasakthi@edss.com                9856734456 23-SEP-24


15 rows selected.
```

3. Find top expensive products.

```
SQL> SELECT * FROM (SELECT * FROM Product ORDER BY Price DESC) WHERE ROWNUM <= 5;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
P1004
Television
SEM0004      75000              5

P1001
Laptops
SEM0001      67500              2

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------

P1006
Tiles
SEM0006      55000            670

P1002
Croptops and shirts

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
```

```
-------- ---------- ---------------
SEM0002       45000             456

P1005
Unruled notes
SEM0005        7500             765
```

4. Display latest orders first.

```
SQL> SELECT * FROM Orders ORDER BY ORDER_DATE DESC;

  ORDER_ID CUSTOMER_ID          ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- -------------------- --------- ------------ --------------------
      1011 C-1005               30-SEP-26        60000 Pending
      1010 C-1005               29-SEP-26         7500 Pending
      1009 C-1004               28-SEP-26         3400 Delivered
      1008 C-1003               27-SEP-26        45000 Shipped
      1007 C-1002               26-SEP-26        67500 Pending
      1006 C-1001               25-SEP-26        55000 Delivered
      1004 C-1004               23-SEP-26        75000 Pending
      1003 C-1003               22-SEP-26         3400 Delivered
      1002 C-1002               21-SEP-26        45000 Shipped
      1001 C-1001               20-SEP-26        70000 Pending

10 rows selected.
```

4.Retrieve unique values using DISTINCT

TASKS:

1.Display unique product categories

```
SQL> SELECT C.CATEGORY_NAME FROM Product P JOIN Category C ON P.CATEGORY_ID = C.CATEGORY_ID;

CATEGORY_NAME
--------------------------------------------------------------------------------
Electronics
Electronics
Electronics
Clothing
Clothing
Clothing
Stationary
Stationary
Stationary
Home Appliances
Home Appliances

CATEGORY_NAME
--------------------------------------------------------------------------------
Mobile Parts
Mobile Parts
Provisions
Provisions

15 rows selected.
```

2. Find different payment methods used by customers.

```
SQL> SELECT DISTINCT PAYMENT_MODE FROM Payment;

PAYMENT_MODE
--------------------
Card
Cash
UPI

SQL>
```

5. Search products based on conditions

TASK:

1. Find products within a specific price range. (Product between 20000 to 40000)
2.Display products belonging to a particular category.

```
SQL> SELECT * FROM Product WHERE PRICE BETWEEN 20000 AND 40000;

no rows selected

SQL> SELECT * FROM Product WHERE CATEGORY_ID = 'SEM0001';

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------
P1001
Laptops
SEM0001      67500              2

P1007
Headphones
SEM0001       2500             25

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY
------- ---------- --------------

P1008
Keyboard
SEM0001       1500             30
```

3.Find products currently available in inventory.

```
SQL> SELECT * FROM Inventory WHERE STOCK_STATUS = 'Available';

INVENTO PRODUCT SELLER_ STOCK_QUANTITY STOCK_STATU LAST_UPDA
------- ------- ------- -------------- ----------- ---------
Inv0001 P1001   S00001               4 Available   12-SEP-26
Inv0002 P1002   S00002               3 Available   13-SEP-26
Inv0004 P1004   S00004               3 Available   14-SEP-26
Inv0005 P1005   S00005              10 Available   16-SEP-26
Inv0006 P1006   S00006              12 Available   17-SEP-26
Inv0007 P1007   S00001              15 Available   18-SEP-26
Inv0008 P1008   S00002              20 Available   19-SEP-26
Inv0009 P1009   S00003               8 Available   20-SEP-26
Inv0010 P1010   S00004               6 Available   21-SEP-26
Inv0011 P1011   S00005              25 Available   22-SEP-26
Inv0012 P1012   S00006              30 Available   23-SEP-26

INVENTO PRODUCT SELLER_ STOCK_QUANTITY STOCK_STATU LAST_UPDA
------- ------- ------- -------------- ----------- ---------
Inv0013 P1013   S00001               7 Available   24-SEP-26
Inv0014 P1014   S00002               9 Available   25-SEP-26
Inv0015 P1015   S00003              14 Available   26-SEP-26

14 rows selected.
```

4.Search products using product name.

5.Find low- stock products.

```
SQL> SELECT * FROM Product WHERE PRODUCT_NAME = 'Laptops';

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGOR       PRICE STOCK_QUANTITY
------- ---------- --------------
P1001
Laptops
SEM0001       67500              2


SQL> SELECT * FROM Inventory WHERE STOCK_QUANTITY < 30;

INVENTO PRODUCT SELLER_ STOCK_QUANTITY STOCK_STATU LAST_UPDA
------- ------- ------- -------------- ----------- ---------
Inv0001 P1001   S00001               4 Available   12-SEP-26
Inv0002 P1002   S00002               3 Available   13-SEP-26
Inv0003 P1003   S00003               4 Unavailable 15-SEP-26
Inv0004 P1004   S00004               3 Available   14-SEP-26
Inv0005 P1005   S00005              10 Available   16-SEP-26
Inv0006 P1006   S00006              12 Available   17-SEP-26
Inv0007 P1007   S00001              15 Available   18-SEP-26
Inv0008 P1008   S00002              20 Available   19-SEP-26
Inv0009 P1009   S00003               8 Available   20-SEP-26
Inv0010 P1010   S00004               6 Available   21-SEP-26
Inv0011 P1011   S00005              25 Available   22-SEP-26

INVENTO PRODUCT SELLER_ STOCK_QUANTITY STOCK_STATU LAST_UPDA
------- ------- ------- -------------- ----------- ---------
Inv0013 P1013   S00001               7 Available   24-SEP-26
Inv0014 P1014   S00002               9 Available   25-SEP-26
Inv0015 P1015   S00003              14 Available   26-SEP-26

14 rows selected.
```

6.Retrieve customer and product information.

TASKS:

1.Display customer details with their orders.

```
SQL> SELECT C.CUSTOMER_ID, C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME, C.EMAIL, C.PHONE, O.ORDER_ID,
O.ORDER_DATE, O.TOTAL_AMOUNT, O.ORDER_STATUS FROM Customer C JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID OR
DER BY C.CUSTOMER_ID, O.ORDER_DATE;

CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
EMAIL                          PHONE   ORDER_ID ORDER_DAT TOTAL_AMOUNT
------------------------------ ---------- ---------- --------- ------------
ORDER_STATUS
--------------------
C-1001               Beema ponting
beema200@edss.com              9234567856       1001 20-SEP-26        70000
Pending

C-1001               Beema ponting
beema200@edss.com              9234567856       1006 25-SEP-26        55000
Delivered

CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
EMAIL                          PHONE   ORDER_ID ORDER_DAT TOTAL_AMOUNT
------------------------------ ---------- ---------- --------- ------------
ORDER_STATUS
--------------------
```

```
C-1002               priya sakthi
priyasakthi@edss.com           9856734456       1002 21-SEP-26        45000
Shipped

C-1002               priya sakthi
priyasakthi@edss.com           9856734456       1007 26-SEP-26        67500

CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
EMAIL                          PHONE   ORDER_ID ORDER_DAT TOTAL_AMOUNT
------------------------------ ---------- ---------- --------- ------------
ORDER_STATUS
--------------------
Pending

C-1003               Arun kumar
arun200@edss.com               9876543212       1003 22-SEP-26         3400
Delivered

C-1003               Arun kumar

CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
EMAIL                          PHONE   ORDER_ID ORDER_DAT TOTAL_AMOUNT
------------------------------ ---------- ---------- --------- ------------
ORDER_STATUS
--------------------
arun200@edss.com               9876543212       1008 27-SEP-26        45000
Shipped

C-1004               Anu Priya
anupriya200@edss.com           8765943256       1004 23-SEP-26        75000
Pending


CUSTOMER_ID          CUSTOMER_NAME
```

```
--------------------
C-1004                Anu Priya
anupriya200@edss.com           8765943256       1009 28-SEP-26         3400
Delivered

C-1005                Reena das
reenadas200@edss.com           9087654567       1010 29-SEP-26         7500
Pending

CUSTOMER_ID           CUSTOMER_NAME
--------------------- ----------------------------------------
EMAIL                                PHONE   ORDER_ID ORDER_DAT TOTAL_AMOUNT
------------------------------ ---------- ---------- --------- ------------
ORDER_STATUS
--------------------

C-1005                Reena das
reenadas200@edss.com           9087654567       1011 30-SEP-26        60000
Pending


10 rows selected.
```

2.Display product details with category information

```
SQL> SELECT P.PRODUCT_ID, P.PRODUCT_NAME, P.PRICE, P.STOCK_QUANTITY, C.CATEGORY_NAME FROM Product P JOIN Categ
ory C ON P.CATEGORY_ID = C.CATEGORY_ID;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
CATEGORY_NAME
--------------------------------------------------------------------------------
P1008
Keyboard
      1500             30
Electronics
```

```
P1007
Headphones
       2500                 25
Electronics


PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
      PRICE STOCK_QUANTITY
---------- --------------
CATEGORY_NAME
--------------------------------------------------------------------------------
P1001
Laptops
      67500                  2
Electronics


PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
      PRICE STOCK_QUANTITY
---------- --------------
CATEGORY_NAME
--------------------------------------------------------------------------------
P1010
Jeans
       1800                 20
Clothing
```

```
----------------------------------------------------------------
P1009
T-Shirts
       800                40
Clothing

PRODUCT
-------
PRODUCT_NAME
----------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- ---------------
CATEGORY_NAME
----------------------------------------------------------------
P1002
Croptops and shirts
     45000               456
Clothing

PRODUCT
-------
PRODUCT_NAME
----------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- ---------------
CATEGORY_NAME
----------------------------------------------------------------
P1012
Pens
        20               200
Stationary

PRODUCT
-------
```

```
----------------------------------------------------------------
P1011
Notebooks
       100                100
Stationary


PRODUCT
-------
PRODUCT_NAME
----------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- ---------------
CATEGORY_NAME
----------------------------------------------------------------
P1003
Fish tank
      3400                 4
Stationary


PRODUCT
-------
PRODUCT_NAME
----------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- ---------------
CATEGORY_NAME
----------------------------------------------------------------
P1013
Mixer Grinder
      3500                15
Home Appliances
```

```
P1004
Television
     75000                5
Home Appliances

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
      PRICE STOCK_QUANTITY
---------- ---------------
CATEGORY_NAME
--------------------------------------------------
P1014
Mobile Charger
       600               35
Mobile Parts

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
      PRICE STOCK_QUANTITY
---------- ---------------
CATEGORY_NAME
--------------------------------------------------
P1005
Unruled notes
      7500              765
Mobile Parts
```

```
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
      PRICE STOCK_QUANTITY
---------- ---------------
CATEGORY_NAME
--------------------------------------------------
P1015
Rice Bag
      1200               50
Provisions

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
      PRICE STOCK_QUANTITY
---------- ---------------
CATEGORY_NAME
--------------------------------------------------
P1006
Tiles
     55000              670
Provisions

15 rows selected.
```

3.Find customers who purchased a specific product

```
SQL> SELECT DISTINCT C.CUSTOMER_ID, C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME FROM Customer C JOIN O
rders O ON C.CUSTOMER_ID = O.CUSTOMER_ID JOIN Order_Details OD ON O.ORDER_ID = OD.ORDER_ID JOIN Product P ON O
D.PRODUCT_ID = P.PRODUCT_ID WHERE P.PRODUCT_NAME = 'Laptops';

CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
C-1002               priya sakthi
C-1001               Beema ponting
```

4.Display products purchased by each customer

```
SQL> SELECT C.CUSTOMER_ID, C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME, P.PRODUCT_ID, P.PRODUCT_NAME F
ROM Customer C JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID JOIN Order_Details OD ON O.ORDER_ID = OD.ORDER_I
D JOIN Product P ON OD.PRODUCT_ID = P.PRODUCT_ID ORDER BY C.CUSTOMER_ID;

CUSTOMER_ID          CUSTOMER_NAME                            PRODUCT
-------------------- ---------------------------------------- -------
PRODUCT_NAME
--------------------------------------------------------------------------------
C-1001               Beema ponting                            P1006
Tiles

C-1001               Beema ponting                            P1001
Laptops

C-1002               priya sakthi                             P1001
Laptops


CUSTOMER_ID          CUSTOMER_NAME                            PRODUCT
-------------------- ---------------------------------------- -------
PRODUCT_NAME
--------------------------------------------------------------------------------
C-1002               priya sakthi                             P1002
Croptops and shirts
```

```
CUSTOMER_ID          CUSTOMER_NAME                            PRODUCT
-------------------- ---------------------------------------- -------
PRODUCT_NAME
--------------------------------------------------------------------------------
C-1002               priya sakthi                             P1002
Croptops and shirts

C-1003               Arun kumar                               P1002
Croptops and shirts

C-1003               Arun kumar                               P1003
Fish tank


CUSTOMER_ID          CUSTOMER_NAME                            PRODUCT
-------------------- ---------------------------------------- -------
PRODUCT_NAME
--------------------------------------------------------------------------------
C-1004               Anu Priya                                P1003
Fish tank

C-1004               Anu Priya                                P1004
Television

C-1005               Reena das                                P1005
Unruled notes


9 rows selected.
```

7.Apply multiple filtering conditions

TASKS:

1.Find electronics products costing more than ₹10000

```
SQL> SELECT P.PRODUCT_ID, P.PRODUCT_NAME, C.CATEGORY_NAME, P.PRICE FROM Product P JOIN Category C ON P.CATEGOR
Y_ID = C.CATEGORY_ID WHERE C.CATEGORY_NAME = 'Electronics' AND P.PRICE > 10000 ORDER BY P.PRICE DESC;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME
--------------------------------------------------------------------------------
     PRICE
----------
P1001
Laptops
Electronics
     67500
```

2. Search products containing the word "mobile"

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE, STOCK_QUANTITY FROM Product WHERE UPPER(PRODUCT_NAME) LIKE '%MOBI
LE%';

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
P1014
Mobile Charger
       600             35
```

3.Find orders placed within a specific date range

```
SQL> SELECT ORDER_ID, CUSTOMER_ID, ORDER_DATE, TOTAL_AMOUNT, ORDER_STATUS FROM Orders WHERE ORDER_DATE BETWEEN
 DATE '2026-09-01' AND DATE '2026-09-30' ORDER BY ORDER_DATE;

  ORDER_ID CUSTOMER_ID          ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- -------------------- --------- ------------ --------------------
      1001 C-1001               20-SEP-26        70000 Pending
      1002 C-1002               21-SEP-26        45000 Shipped
      1003 C-1003               22-SEP-26         3400 Delivered
      1004 C-1004               23-SEP-26        75000 Pending
      1006 C-1001               25-SEP-26        55000 Delivered
      1007 C-1002               26-SEP-26        67500 Pending
      1008 C-1003               27-SEP-26        45000 Shipped
      1009 C-1004               28-SEP-26         3400 Delivered
      1010 C-1005               29-SEP-26         7500 Pending
      1011 C-1005               30-SEP-26        60000 Pending

10 rows selected.
```

8. Generate basic business reports

Report 1: product availability report

Display:

- Product name
- Price
- stock quantity
- availability status

```
SQL> SELECT P.PRODUCT_NAME, P.PRICE, I.STOCK_QUANTITY, I.STOCK_STATUS FROM Product P JOIN Inventory I ON P.PRODUCT_ID = I.PRODUCT_ID;

PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY STOCK_STATU
---------- -------------- -----------
Laptops
     67500              4 Available

Croptops and shirts
     45000              3 Available

Fish tank
      3400              4 Unavailable

PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY STOCK_STATU
---------- -------------- -----------
Television
     75000              3 Available

Unruled notes
      7500             10 Available

Tiles
     55000             12 Available

PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY STOCK_STATU
---------- -------------- -----------
```

```
       2500                15 Available

Keyboard
       1500                20 Available

T-Shirts
        800                 8 Available


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY STOCK_STATU
---------- -------------- -----------
Jeans
       1800                 6 Available

Notebooks
        100                25 Available

Pens
         20                30 Available


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY STOCK_STATU
---------- -------------- -----------
Mixer Grinder
       3500                 7 Available

Mobile Charger
        600                 9 Available

Rice Bag
       1200                14 Available


15 rows selected.
```

Report 2: customer report generate:

Display:

- Total number of customers

```
SQL> SELECT COUNT(*) AS TOTAL_CUSTOMERS FROM Customer;

TOTAL_CUSTOMERS
---------------
             15
```

- New customer registrations

```
SQL> SELECT CUSTOMER_ID, FIRST_NAME || ' ' || LAST_NAME AS CUSTOMER_NAME, EMAIL, CREATED_AT FROM Customer WHERE CREATED_AT >= DATE '2026-01-01'
AND CREATED_AT < DATE '2027-01-01' ORDER BY CREATED_AT DESC;

CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
EMAIL                          CREATED_A
------------------------------ ---------
C-1015               Nithin Kumar
nithin115@gmail.com            16-MAR-26

C-1014               Sneha Rani
sneha114@gmail.com             09-FEB-26

C-1013               Arjun Raj
arjun113@gmail.com             14-JAN-26
```

Report 3: Order Report

Display:

- Total Orders

```
SQL> SELECT COUNT(*) AS TOTAL_ORDERS
  2  FROM ORDERS;

TOTAL_ORDERS
------------
          10
```

- Completed orders

```
SQL> SELECT COUNT(*) AS COMPLETED_ORDERS
  2  FROM ORDERS
  3  WHERE ORDER_STATUS = 'Delivered';

COMPLETED_ORDERS
----------------
               3
```

- Pending orders

```
SQL> SELECT COUNT(*) AS PENDING_ORDERS
  2  FROM ORDERS
  3  WHERE ORDER_STATUS IN ('Processing','Shipped');

PENDING_ORDERS
--------------
             2
```

- Cancelled orders

```
SQL> SELECT COUNT(*) AS CANCELLED_ORDERS
  2  FROM ORDERS WHERE ORDER_STATUS = 'Cancelled';

CANCELLED_ORDERS
----------------
               0
```

Report 4: Product Performance Report

Find:

- Highest-priced products

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE FROM Product ORDER BY PRICE DESC;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
P1004
Television
     75000

P1001
Laptops
     67500

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

P1006
Tiles
     55000

P1002
Croptops and shirts

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
```

```
-----------
      45000

P1005
Unruled notes
       7500

P1013

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
      PRICE
-----------
Mixer Grinder
       3500

P1003
Fish tank
       3400


PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
      PRICE
-----------
P1007
Headphones
       2500

P1010
Jeans
       1800
```

```
P1008
Keyboard
       1500

P1015
Rice Bag

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
      PRICE
-----------
       1200

P1009
T-Shirts
        800

P1014

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------
      PRICE
-----------
Mobile Charger
        600

P1011
Notebooks
        100
```

```
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
P1012
Pens
        20


15 rows selected.
```

- Most reviewed products

```
SQL> SELECT P.PRODUCT_ID, P.PRODUCT_NAME, COUNT(R.REVIEW_ID) AS REVIEW_COUNT FROM Product P JOIN Review R ON P.PRODUCT_ID = R.PRODUCT_ID GROUP B
Y P.PRODUCT_ID, P.PRODUCT_NAME ORDER BY REVIEW_COUNT DESC;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
REVIEW_COUNT
------------
P1001
Laptops
         2

P1002
Croptops and shirts
         2

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
REVIEW_COUNT
------------

P1004
Television
         2

P1003
Fish tank

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
REVIEW_COUNT
```

```
REVIEW_COUNT
------------
               1

P1005
Unruled notes
               1

P1006

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
REVIEW_COUNT
------------
Tiles
               1


6 rows selected.
```

- Available products

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE, STOCK_QUANTITY FROM Product WHERE STOCK_QUANTITY > 0 ORDER BY STOCK_QUANTITY DESC;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
P1005
Unruled notes
      7500            765

P1006
Tiles
     55000            670

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------

P1002
Croptops and shirts
     45000            456

P1012
Pens

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
```

```
          20                200

P1011
Notebooks
         100                100

P1015

PRODUCT
--------
PRODUCT_NAME
--------------------------------------------------------------------------------
      PRICE  STOCK_QUANTITY
-----------  ----------------
Rice Bag
        1200                50

P1009
T-Shirts
         800                40


PRODUCT
--------
PRODUCT_NAME
--------------------------------------------------------------------------------
      PRICE  STOCK_QUANTITY
-----------  ----------------
P1014
Mobile Charger
         600                35

P1008
Keyboard
        1500                30
```

```
P1007
Headphones
      2500              25

P1010
Jeans

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
      1800              20

P1013
Mixer Grinder
      3500              15

P1004

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
Television
     75000               5

P1003
Fish tank
      3400               4
```

```
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
P1001
Laptops
     67500               2


15 rows selected.
```